

Math 118 Lec 7

- plan:
- use of non-orthonormal bases (examples)
 - Gram-Schmidt via the Cholesky factorization of G
 - least squares

recap: Suppose $\varphi_1, \dots, \varphi_N \in V$ are linearly independent

and define $T: \mathbb{C}^N \rightarrow V$

by

$$Tc = \sum_{j=1}^N c_j \varphi_j$$

Let $V_0 = \text{ran}(T) = \text{span}\{\varphi_1, \dots, \varphi_N\}$

1. The Gram matrix $G = T^*T$ ($G_{jk} = \langle \varphi_k, \varphi_j \rangle$) is invertible.

2. The orthogonal projection P of V onto V_0 is given by

$$P = T G^{-1} T^*$$

Notation: when $\mathbb{C}^N$ or $\mathbb{R}^N$ is involved (here it's the domain of T) then e_k usually denotes the elementary unit vectors there ($(e_k)_j = \delta_{kj}$)

So the Gram-Schmidt vectors obtained from $\varphi_1, \dots, \varphi_N$ will instead be denoted by $\psi_1, \dots, \psi_N$.

Also, sometimes it's better to start from $j=0$.

Example: find the closest point to $f(x) = \sin(\pi x)$ in the subspace $V_0 = \text{span}\{1, x, x^2\}$ in $L^2(0,1)$.

option 1: do Gram-Schmidt to get ψ_0, ψ_1, ψ_2 O.N.

Then compute $c_k = \langle f, \psi_k \rangle$, $Pf(x) = \sum_{k=0}^2 c_k \psi_k$

option 2: $(T^*T)_{jk} = \langle \varphi_k, \varphi_j \rangle = \int_0^1 x^k x^j dx = \frac{x^{k+j+1}}{k+j+1} \Big|_0^1 = \frac{1}{k+j+1}$

$$(T^*f)_j = \langle f, \varphi_j \rangle = \int_0^1 (\sin \pi x) x^j dx$$

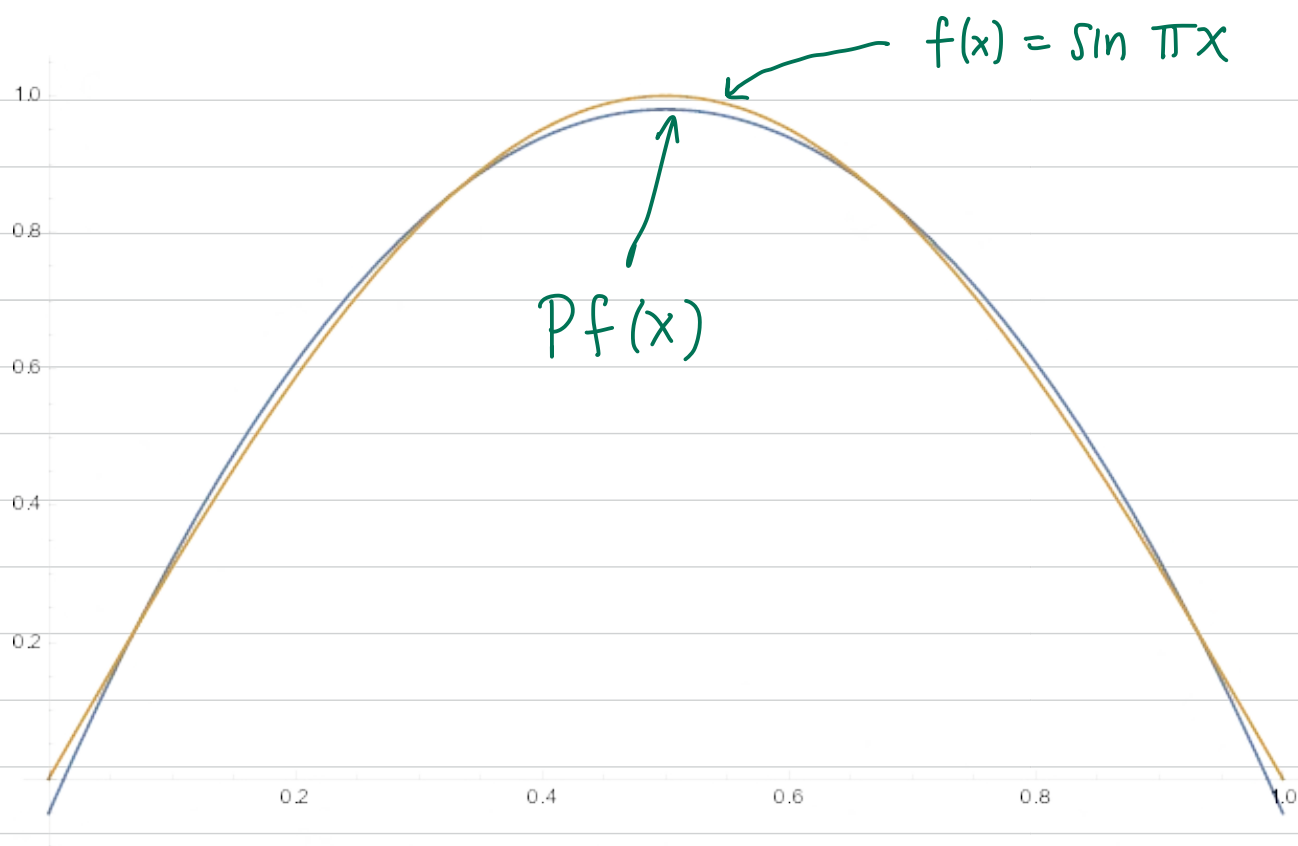
$$j=0: \int_0^1 \sin \pi x dx = -\frac{\cos \pi x}{\pi} \Big|_0^1 = \frac{2}{\pi}$$

$$j=1: \int_0^1 x \sin \pi x dx = \left[-x \frac{\cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_0^1 = \frac{1}{\pi}$$

$$j=2: \int_0^1 x^2 \sin \pi x dx = \left[-x^2 \frac{\cos \pi x}{\pi} + 2x \frac{\sin \pi x}{\pi^2} + 2 \frac{\cos \pi x}{\pi^3} \right]_0^1 = \frac{1}{\pi} + 0 - \frac{4}{\pi^3}$$

$$Pf(x) = (1, x, x^2) \begin{pmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{pmatrix}^{-1} \begin{pmatrix} 2/\pi \\ 1/\pi \\ \frac{\pi^2-4}{\pi^3} \end{pmatrix}$$

$$= \frac{12}{\pi^3} (\pi^2 - 10) + \frac{60}{\pi^3} (12 - \pi^2) x - \frac{60}{\pi^3} (12 - \pi^2) x^2$$



How is option 2 related to Gram-Schmidt?

$$P = T G^{-1} T^*, \quad G = T^* T$$

Compute R so that $G = R^* R$

matlab command: $R = \text{chol}(G)$

Mathematica " : $R = \text{CholeskyDecomposition}(G)$

Cholesky factorization

R upper triangular

R^* = conjugate transpose

$r_{kk} > 0$ (positive diagonal entries)

theorem: every positive definite matrix has a Cholesky factorization.

question in class: how do you compute R ?

answer: first compute the LU factorization: $A = LU$.

procedure: (row reduction) $M_{n-1} \cdots M_2 M_1 A = U$

left multiplication by $M_j = I - \tau_j e_j^T$

performs elementary row operations to cancel the entries in the j th column below the main diagonal

$$\begin{array}{l} A^{(1)} = A \\ M_1 A^{(1)} = A^{(2)} \\ M_2 A^{(2)} = A^{(3)} \\ \vdots \\ M_{n-1} A^{(n-1)} = A^{(n)} = U \end{array} \quad \left| \quad \begin{array}{l} \text{e.g. when } n=4 \\ M_2 A^{(2)} = A^{(3)} \text{ has the form} \\ \begin{pmatrix} 1 & & & \\ & 1 & & \\ -l_{32} & & 1 & \\ -l_{42} & & & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & & & \\ 0 & \{a_{ij}^{(2)}\}_{2 \leq i \leq 4, 2 \leq j \leq 4} & & \\ 0 & & & \end{pmatrix} \end{array} \right.$$

$$\begin{array}{l} \text{zero out} \\ a_{32}^{(3)}, a_{42}^{(3)} \end{array} \rightarrow \begin{cases} l_{32} = a_{32}^{(2)} / a_{22}^{(2)} \\ l_{42} = a_{42}^{(2)} / a_{22}^{(2)} \end{cases}$$

$$\text{here } \tau_2 = \begin{pmatrix} 0 \\ 0 \\ l_{32} \\ l_{42} \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, M_2 = I - \tau_2 e_2^T$$

Then update the other entries of $M_2 A^{(2)}$ to get $A^{(3)}$

In the end, $M_{n-1} \cdots M_1 A = U$ is upper triangular

$$\text{and } \boxed{A = M_1^{-1} \cdots M_{n-1}^{-1} U = LU}$$

Cool facts: ① $M_j^{-1} = I + \tau_j e_j^T$
 just change the sign

$$② L_{ij} = \begin{cases} 0 & i < j \\ 1 & i = j \\ l_{ij} & i > j \end{cases}$$

the same numbers l_{ij} from the cancellation formula: $l_{ij} = \frac{a_{ij}^{(j)}}{a_{jj}^{(j)}}$

Once you have $A = LU$,

$$\text{let } D = \text{diag}(U) = \begin{pmatrix} u_{11} & & 0 \\ & u_{22} & \\ 0 & & \ddots & \\ & & & u_{nn} \end{pmatrix}$$

When A is symmetric positive definite (SPD)

each entry u_{kh} is positive, and $A = L \underbrace{D L^*}_U$

$$A = L D^{1/2} D^{-1/2} U = R^* R$$

Where $R = D^{-1/2} U$, $L D^{1/2} = R^*$ added after class to answer the question

Now I am switching to $U = R^{-1}$ (since U is a generic symbol for an upper triangular matrix)

R^{-1} is also upper triangular (can backsolve $R U = I$)

$u_k = k\text{th column of } U = R^{-1}$
 satisfies $\underbrace{(u_k)_j}_{\text{or } u_{jk}} = 0 \text{ for } j \geq k+1$

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ & \cdot & \cdot & \cdot & \cdot \\ & & \cdot & \cdot & \cdot \\ & & & \cdot & \cdot \\ & & & & \cdot \end{pmatrix} \begin{pmatrix} * \\ * \\ * \\ * \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \leftarrow e_k = k\text{th column of } I$$

(zeros at end) ($(e_k)_j = 0$ for $j \geq k+1$)

$$\begin{aligned}
 P &= T G^{-1} T^* = T (R^* R)^{-1} T^* \\
 &= T R^{-1} (R^*)^{-1} T^* = (T R^{-1}) (T R^{-1})^* \\
 &= \Psi \Psi^*
 \end{aligned}$$

$$\begin{aligned}
 \Psi^* \Psi &= (T R^{-1})^* (T R^{-1}) \\
 &= (R^{-1})^* T^* T R^{-1}
 \end{aligned}$$

$$= (R^*)^{-1} R^* R R^{-1} = (I)(I) = I$$

$$Pf = \sum_{k=1}^N \langle f, \psi_k \rangle \psi_k$$

Claim: $\Psi = T R^{-1}$ is the same orthonormal basis matrix that you get from doing Gram-Schmidt on $\varphi_1, \dots, \varphi_N$

$$(\psi_1, \dots, \psi_N) = (\varphi_1, \dots, \varphi_N) \begin{pmatrix} u_{11} & u_{12} & \dots & u_{1N} \\ & u_{22} & & \vdots \\ & & \ddots & \\ & & & u_{NN} \end{pmatrix}$$

$U = R^{-1} \rightarrow$

$$\langle \psi_k, \psi_j \rangle = \langle \Psi e_k, \Psi e_j \rangle = \langle \underbrace{\Psi^* \Psi}_I e_k, e_j \rangle = \delta_{kj}$$

and, because U is upper triangular, ψ_k is a linear combination of $\varphi_1, \dots, \varphi_k$ (same as Gram-Schmidt)

$$(\varphi_{kk} > 0 \Rightarrow u_{kk} > 0, \text{ so } \psi_k = \underbrace{u_{kk}}_{\text{positive (same as Gram-Schmidt)}} \varphi_k + \sum_{j=1}^{k-1} u_{jk} \varphi_j)$$

Least squares.

$$A: \mathbb{C}^n \rightarrow \mathbb{C}^m, \quad m \geq n$$

A $m \times n$ matrix, $b \in \mathbb{C}^m$

$$\begin{matrix} & n \\ & \boxed{A} \\ m & \end{matrix} \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} b \end{bmatrix}$$

If $m > n$, the system is overdetermined
(usually $Ax=b$ has no solution,
except in the least squares sense)

Goal: find x to minimize $\|Ax-b\|_2$

$$V_0 = \text{range}(A) = \text{span}(\varphi_1, \dots, \varphi_N), \quad \varphi_k = k\text{th col of } A$$

We're looking for the closest point in V_0 to b

solution: $Pb = \underbrace{A(A^*A)^{-1}A^*b}_x$

$$A^*A x = A^*b \quad \leftarrow \text{normal equations}$$

$$\left(\begin{array}{l} A^*A \text{ is invertible iff} \\ \varphi_1, \dots, \varphi_N \text{ are linearly independent} \end{array} \right)$$

Example: best-fit lines and curves

x will have another meaning, so we will
be solving $A \overset{LS}{c} = y$, i.e. minimizing $\|y - Ac\|$

Setup: given data points $(x_1, y_1), \dots, (x_m, y_m)$
and functions $\varphi_1(x), \dots, \varphi_n(x)$

find $c_1, \dots, c_n$ to minimize $\sum_{i=1}^m |y_i - \sum_{j=1}^n c_j \varphi_j(x_i)|^2$

$Tc = T(x)c = \sum_{j=1}^n c_j \varphi_j(x)$ is the best-fit curve

solution: construct the system $A_{ij} = \varphi_j(x_i)$

solve $Ac = y$ in the least-squares sense.

$$A = \begin{pmatrix} \varphi_1(x_1) & \dots & \varphi_n(x_1) \\ \vdots & & \vdots \\ \varphi_1(x_m) & \dots & \varphi_n(x_m) \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}.$$

$A^*Ac = A^*y$
 $\uparrow$
 $n \times n$ linear system.

e.g. the best fit line to

x	0	1	1	2
y	0	0	1	2

is found via $\varphi_0(x) = 1, \varphi_1(x) = x \leftarrow c_0 = b, c_1 = m$

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad y = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

fit function: $T(x)c = mx + b$

$$A^*Ac = A^*y$$

$$A^T A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 6 \end{pmatrix} \quad \left| \quad \begin{pmatrix} 4 & 4 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} b \\ m \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \right.$$

$$A^T y = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \quad \left| \quad \begin{pmatrix} 4 & 4 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} b \\ m \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right.$$

$$m=1, 4b = 3-4 = -1$$

$T(x)C = X - \frac{1}{4}$

There are better alternatives for solving least squares problems than via the normal equations. (QR factorization, SVD)

return to " $Ax = b$ " notation.

method 1: normal equations

method 2: QR factorization

upper triangular
and invertible



find Q $m \times m$, $Q^* Q = I_{m \times m}$, $A = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$

since Q is orthogonal ($Q^* Q = I$) it preserves lengths (see HW3)

$$\|Q^* y\| = \|y\| \quad \forall y \in \mathbb{C}^m$$

$$\|Ax - b\|_2 = \|Q^*(Ax - b)\|_2$$

$$Q^* A = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

matlab command: $[Q, R] = \text{qr}(A)$

mathematica: $\{Q1, R\} = \text{QRDecomposition}(A)$
(returns $Q1 = Q^*$)

then $\begin{bmatrix} R \\ 0 \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = Q^* b = \begin{bmatrix} y \\ r \end{bmatrix}$

← can't do anything about these components, so minimizing the residual is equivalent to zeroing out $y = \text{"first } n \text{ components of } Q^* b \text{"}$.

solution is $x = R^{-1} y$

Yet another approach is via the singular value decomposition of A ,

$A = U S V^*$

matlab: $[U, S, V] = \text{svd}(A)$

mathematica: $\{U, S, V\} = \text{SingularValueDecomposition}(A)$

$$U \text{ } m \times m, V \text{ } n \times n, U^* U = I, V^* V = I$$

$$S = \begin{pmatrix} \Sigma \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{pmatrix}, \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$$

$$Ax = b$$

$$U S V^* x = b$$

$$\begin{pmatrix} \Sigma \\ 0 \end{pmatrix} V^* x = U^* b = \begin{pmatrix} y \\ r \end{pmatrix}$$

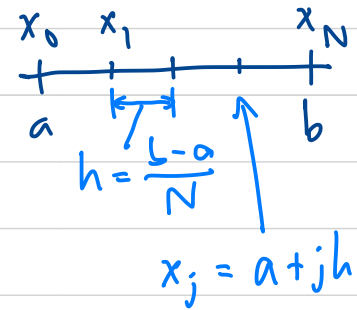
solution is $x = V \Sigma^{-1} y$

again, can't do anything about r

Additional example where non-orthonormal bases are useful.

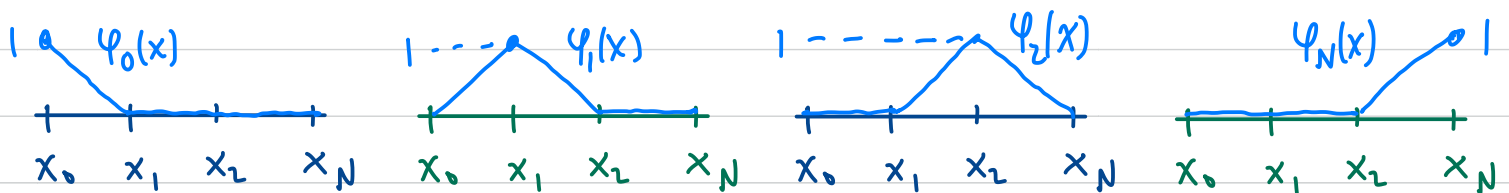
(supplementary. Omit in class)

Let V_0 be the set of piecewise linear functions with equally spaced knots.



Find the closest function $u_0 \in V_0$ to a given function $u \in L^2(a, b)$

Solution: Let $\{\varphi_k\}_{k=0}^N$ be a nodal basis for V_0 ($\varphi_k(x_j) = \delta_{kj}$)



The Gram matrix $(T^T T)_{jk} = \langle \varphi_k, \varphi_j \rangle$ is tridiagonal ("element by element assembly")

$$\langle \varphi_j, \varphi_{j+1} \rangle = \int_0^h \left(1 - \frac{x}{h}\right) \left(\frac{x}{h}\right) dx = \frac{x^2}{2h} - \frac{x^3}{3h} \Big|_0^h = \frac{h}{6}$$

$$\langle \varphi_j, \varphi_j \rangle = \int_{x_{j-1}}^{x_j} \left(\frac{x}{h}\right)^2 dx + \int_{x_j}^{x_{j+1}} \left(1 - \frac{x}{h}\right)^2 dx = \left\{ \begin{aligned} &\int_0^h \left(\frac{x}{h}\right)^2 dx = \frac{x^3}{3h^2} \Big|_0^h = \frac{h}{3} \\ &+ \int_0^h \left(1 - \frac{x}{h}\right)^2 dx = -\frac{(h-x)^3}{3h^2} \Big|_0^h = \frac{h}{3} \end{aligned} \right.$$

$j=N$: only include this one

$j=0$: only include this one

result:

$$u_0 = Pu = \underbrace{(\quad)}_{T(T^*T)^{-1}T^*u} = (\varphi_0, \dots, \varphi_N)$$

$$\begin{pmatrix} h/3 & h/6 & & & \\ h/6 & 2h/3 & h/6 & & \\ & h/6 & 2h/3 & \ddots & \\ & & \ddots & h/6 & 2h/3 & h/6 \\ & & & h/6 & h/3 \end{pmatrix}^{-1} \begin{pmatrix} \langle u, \varphi_0 \rangle \\ \vdots \\ \langle u, \varphi_N \rangle \end{pmatrix}$$

key point: T^*T is sparse with this choice of basis
($\Rightarrow$ cheap to compute $(T^*T)^{-1}(T^*u)$)

Using Gram-Schmidt instead would be much more expensive.

(These ideas are at the heart of the finite element method for solving PDEs)